TITLE: Phan Tuấn Kiệt 2352654 Date: HW5

Activity 1:

For example in problem moving from the top left corner to the right bottom corner. There are four possible moving (~~up~~ up, down, left, right) and the maximum move is 12. 0 1 2 3

I can describe ~~how~~ the way to move by using an array with 12 element, ~~each~~ each elements $\in [0; 3]$ represent for direction.

~~I can calculate the result~~

I can evaluate how good the path by calculating distance ~~from~~ between ~~desti~~ goal and ~~destination~~ of way

Because the requirement is move from top left corner to bottom right corner so that the unexpected chromosome ~~for~~ we do not ~~have~~ any move down or move right.

to deal this problem.

I can hybrid this chromosome with another chromosome.

The I mutation ~~have~~ DNA of child to make sure that the child donot ~~have~~ inherit all bad feature from parent.

Activity 2:

$$F(2) = \frac{Accuracy \cdot N}{num\ of\ feature\ chosen}$$ (num of feature chosen $\in N^*$)

I choose this function because the requirement penalty ~~to~~ the model ~~with~~ that which use the lot of feature → number of chosen feature inversely proportional with fitness ~~while the accuracy reduce prop~~ while accuracy proportional with fitness. I multi with N (number of features) just only ensure the number not too male.

Activity 3:

10100 10111 00 —mutation→ 10110 10 10100

10110 10|10100 → 10110 10|0001

0 01,00|10001 → 00 1 00|10100

in permutation it very hard to use generic algorithm because the value at ~~each~~ first index has n possible value elements, the next index have n-1 index

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possible value and so on. Since an element cannot be repeated. This constraint on chromosome generation makes mutation and crossover difficult, as these operations risk violating the permutations rules.

Activity 4:

First, mutation rate ~~was~~ quite ~~high~~ low, it leading to the population saturation following crossover operations, resulting in a lack of diversity within the population. Secondly, consistently selecting the individual with the highest fitness value causes the entire population to eventually inherit only that individual's traits, halting further evolution.

To deal these issues, the mutation rate should be increased. Additionally, the selection method for breeding needs to be modified. Instead of always choosing the strongest individual in the population, two subsets should be selected at random, and the best individual from each subset chosen for breeding to avoid inbreeding.

Activity 5.

The chromosome in my code represented as a binary array with a length equal to the total number of data features.

The fitness function used is: Fitness = Accuracy — penalty. The penalty is calculated by 0,001 x number of selected features while accuracy is determin by Gaussian NB model. The population size is set to 20 and the stopping condition is reached 15 generation. Three seed were used [31, 153, 2021].

The result obtain:

Seed 31 : fitness: 0,5292

Seed 153 : fitness: 0,6184

Seed 2021 : fitness: 0,5546.

The result from the three seeds reveal significant variation. Seed 153 achieved a fitness = 0,6184, while seed 31 get the lower value of 0,5292. This discrepancy arises because the genetic algorithm relies on randomness at several stages (initial population selection, crossover selection, mutation probability) leading to instability across training runs, even when using the same dataset.

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